

Supplementary Material

for

Physics-Constrained Multisensor Bayesian Inversion of Hydrologic Storage from InSAR Deformation

Mehdi Joud

April 19, 2026

This document provides the supplementary appendices accompanying the main manuscript.

The main paper presents the compact methodology, datasets, grouped multisensor real-data inversion, and principal validation results. This supplementary document contains the detailed mathematical proofs, forward formulations, conditioning analysis, and extended validation material that support the main manuscript.

Contents

A	Bayesian Feasibility of the Constrained Deformation-Space Prior	3
A.1	Functional setting and notation	3
A.2	Forward deformation operator	3
A.3	Linear-Gaussian state-space model	4
A.4	Existence and uniqueness of the posterior	5
A.5	Posterior mean and MAP estimator	5
A.6	Bayes-optimality under squared loss	6
A.7	Induced posterior estimate for the state	6
A.8	What cannot be claimed: non-identifiability of the full five-layer state	6
A.9	Constrained extension	7
A.10	Real-data conditioning and the grouped Bologna formulation	7
B	Forward Hydro-Mechanical Formulation and Observation Operators	8
B.1	Full five-layer hydrologic state	8
B.2	Continuous forward deformation model	8
B.3	LOS projection and deformation observation	9
B.4	Discrete operator form	9
B.5	Grouped state used in the applied Bologna inversion	10
B.6	Regional aggregation operators	10
B.7	GRACE observation operator	10
B.8	SMAP observation operator	11
B.9	SWOT observation operator	11
B.10	Grouped multisensor observation model	11
B.11	State evolution in the grouped Bologna model	11
B.12	Interpretation	12

C	Conditioning Diagnostics of the Bologna Inversion	12
C.1	Purpose	12
C.2	Inverse-problem setting	12
C.2.1	Old deformation-space Stage 1 formulation	12
C.2.2	Current grouped balanced multisensor Stage 1	13
C.3	Conditioning metrics	13
C.3.1	Column-scale spread	13
C.3.2	Singular values and operator condition number	14
C.3.3	Thresholded numerical rank	14
C.3.4	Information matrix / Gauss–Newton Hessian	14
C.3.5	Column correlation / separability	15
C.3.6	Observability Gramian	15
C.4	How the diagnostics were computed from the saved runs	15
C.4.1	Old layered and grouped patch runs	15
C.4.2	Current grouped multisensor Stage 1	16
C.5	Result tables	16
C.6	Interpretation	16
D	Synthetic Validation and External Well-Validation Details	17
D.1	Synthetic validation protocol	17
D.2	External well-validation methodology	18
D.3	Overall well-validation results	19
D.4	Depth dependence and best-state interpretation	20
D.5	Trusted validation subset	20
D.6	Interpretation	21

A Bayesian Feasibility of the Constrained Deformation-Space Prior

This appendix states the precise mathematical sense in which the Bayesian core of the proposed framework yields a well-defined and optimal posterior estimate. The conclusions apply to the linear-Gaussian Stage 1 model. They do not imply unique recovery of the full hydrologic truth from deformation alone.

A.1 Functional setting and notation

Let $\Omega \subset \mathbb{R}^2$ denote the spatial domain and let $T \in \mathbb{N}$ be the number of acquisition times. For each $t = 1, \dots, T$, define the hydrologic state

$$\mathbf{x}_t(\mathbf{s}) = \begin{bmatrix} S0_t(\mathbf{s}) \\ Ss_t(\mathbf{s}) \\ Sd_t(\mathbf{s}) \\ Sg_t(\mathbf{s}) \\ Sr_t(\mathbf{s}) \end{bmatrix}, \quad \mathbf{s} \in \Omega. \quad (\text{S1})$$

We define the state space

$$X := (L^2(\Omega))^5, \quad (\text{S2})$$

equipped with the norm

$$\|\mathbf{x}\|_X^2 = \|S0\|_{L^2(\Omega)}^2 + \|Ss\|_{L^2(\Omega)}^2 + \|Sd\|_{L^2(\Omega)}^2 + \|Sg\|_{L^2(\Omega)}^2 + \|Sr\|_{L^2(\Omega)}^2. \quad (\text{S3})$$

The observation space is

$$Y := L^2(\Omega), \quad (\text{S4})$$

with its usual norm.

After discretization on a spatial grid with P cells, the state and observation spaces become

$$X_h = \mathbb{R}^{5P}, \quad Y_h = \mathbb{R}^P. \quad (\text{S5})$$

The time-stacked spaces are

$$\mathcal{X}_h = \mathbb{R}^{5PT}, \quad \mathcal{Y}_h = \mathbb{R}^{PT}. \quad (\text{S6})$$

For each time t , let the discretized W3RA prior state be

$$\mathbf{z}_t \in X_h = \mathbb{R}^{5P}. \quad (\text{S7})$$

We write

$$\mathbf{z}_t = \begin{bmatrix} \mathbf{z}_t^{(S0)} \\ \mathbf{z}_t^{(Ss)} \\ \mathbf{z}_t^{(Sd)} \\ \mathbf{z}_t^{(Sg)} \\ \mathbf{z}_t^{(Sr)} \end{bmatrix}, \quad \mathbf{z}_t^{(\cdot)} \in \mathbb{R}^P. \quad (\text{S8})$$

A.2 Forward deformation operator

Define the continuous linear forward operator

$$\mathcal{A} : X \rightarrow Y \quad (\text{S9})$$

by

$$\mathcal{A}(\mathbf{x}_t) = G_{\text{load}} * (S0_t + Ss_t + Sd_t + Sr_t) + G_{\text{poro}} * Sg_t. \quad (\text{S10})$$

After discretization this becomes

$$A_h : X_h \rightarrow Y_h, \quad A_h \in \mathbb{R}^{P \times 5P}. \quad (\text{S11})$$

To model multiplicative correction of the W3RA prior state, define the diagonal multiplication operator

$$D(\mathbf{z}_t) : \mathbb{R}^{5P} \rightarrow X_h, \quad D(\mathbf{z}_t)\boldsymbol{\theta}_t = \boldsymbol{\theta}_t \odot \mathbf{z}_t, \quad (\text{S12})$$

where $\odot$ denotes elementwise multiplication and

$$\boldsymbol{\theta}_t \in \mathbb{R}^{5P}. \quad (\text{S13})$$

Hence the Stage 1 prior state estimate is

$$\mathbf{x}_t^{\text{prior}} = D(\mathbf{z}_t)\boldsymbol{\theta}_t. \quad (\text{S14})$$

The resulting observation model is therefore

$$\mathbf{y}_t = A_h D(\mathbf{z}_t)\boldsymbol{\theta}_t + \boldsymbol{\varepsilon}_t. \quad (\text{S15})$$

A.3 Linear-Gaussian state-space model

Define

$$H_t := A_h D(\mathbf{z}_t), \quad H_t \in \mathbb{R}^{P \times 5P}. \quad (\text{S16})$$

Then for each $t = 1, \dots, T$, the deformation-space observation equation is

$$\mathbf{y}_t = H_t \boldsymbol{\theta}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, R_t), \quad (\text{S17})$$

where $R_t \in \mathbb{R}^{P \times P}$ is symmetric positive definite.

The temporal evolution is modeled as

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, Q_t), \quad (\text{S18})$$

where $Q_t \in \mathbb{R}^{5P \times 5P}$ is symmetric positive definite.

The initial prior is assumed proper Gaussian:

$$\boldsymbol{\theta}_1 \sim \mathcal{N}(m_0, P_0), \quad P_0 \succ 0. \quad (\text{S19})$$

Assumptions. Throughout this appendix we assume:

1. $R_t \succ 0$ for all t ,
2. $Q_t \succ 0$ for all t ,
3. $P_0 \succ 0$,
4. the discretized model is finite-dimensional.

A.4 Existence and uniqueness of the posterior

Define the stacked latent vector

$$\Theta = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_T \end{bmatrix} \in \mathcal{X}_h = \mathbb{R}^{5PT}, \quad (\text{S20})$$

and the stacked observation vector

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_T \end{bmatrix} \in \mathcal{Y}_h = \mathbb{R}^{PT}. \quad (\text{S21})$$

The linear-Gaussian state dynamics induce a proper Gaussian prior

$$\Theta \sim \mathcal{N}(\mathbf{m}_0, \Sigma_0), \quad \Sigma_0 \succ 0, \quad (\text{S22})$$

and the likelihood is Gaussian:

$$\mathbf{y} \mid \Theta \sim \mathcal{N}(K\Theta, R), \quad (\text{S23})$$

where K is the stacked observation operator and R is block diagonal with blocks R_t .

Proposition 1. *Under the assumptions above, the posterior distribution $\pi(\Theta \mid \mathbf{y})$ exists, is unique, and has finite first and second moments.*

Proof. By Bayes' theorem,

$$\pi(\Theta \mid \mathbf{y}) \propto \exp \left(-\frac{1}{2} \|\mathbf{y} - K\Theta\|_{R^{-1}}^2 - \frac{1}{2} \|\Theta - \mathbf{m}_0\|_{\Sigma_0^{-1}}^2 \right). \quad (\text{S24})$$

Since $R \succ 0$ and $\Sigma_0 \succ 0$, both quadratic forms are coercive. Because $R \succ 0$ and $\Sigma_0 \succ 0$, the matrices R^{-1} and Σ_0^{-1} are positive definite. In particular, the prior term

$$\frac{1}{2} \|\Theta - \mathbf{m}_0\|_{\Sigma_0^{-1}}^2$$

grows quadratically as $\|\Theta\| \rightarrow \infty$, so the negative log-posterior is coercive. Hence the posterior density is integrable and defines a proper Gaussian measure on the finite-dimensional space $\mathcal{X}_h$. Therefore the right-hand side defines a proper Gaussian density on the finite-dimensional space $\mathcal{X}_h$, and hence the posterior exists, is unique, and has finite first and second moments. $\square$

A.5 Posterior mean and MAP estimator

Because the posterior is Gaussian, it has the form

$$\Theta \mid \mathbf{y} \sim \mathcal{N}(\hat{\Theta}, \Sigma_{\text{post}}), \quad (\text{S25})$$

with

$$\Sigma_{\text{post}} = \left(K^\top R^{-1} K + \Sigma_0^{-1} \right)^{-1}, \quad (\text{S26})$$

and

$$\hat{\Theta} = \Sigma_{\text{post}} \left(K^\top R^{-1} \mathbf{y} + \Sigma_0^{-1} \mathbf{m}_0 \right). \quad (\text{S27})$$

Proposition 2. *In the linear-Gaussian Stage 1 model, the posterior mean and the MAP estimator coincide.*

Proof. A Gaussian density attains its unique maximum at its mean. Hence the vector $\hat{\Theta}$ in (S27) is both the posterior mean and the unique maximizer of the posterior density. $\square$

A.6 Bayes-optimality under squared loss

Theorem 1. *Let the loss function be*

$$L_2(\tilde{\Theta}, \Theta) = \|\tilde{\Theta} - \Theta\|_2^2. \quad (\text{S28})$$

Then the Bayes estimator is the posterior mean:

$$\tilde{\Theta}^* = \mathbb{E}[\Theta \mid \mathbf{y}]. \quad (\text{S29})$$

Proof. Let

$$\bar{\Theta} := \mathbb{E}[\Theta \mid \mathbf{y}]. \quad (\text{S30})$$

Then

$$\begin{aligned} \|\tilde{\Theta} - \Theta\|_2^2 &= \|\tilde{\Theta} - \bar{\Theta}\|_2^2 + \|\Theta - \bar{\Theta}\|_2^2 \\ &\quad - 2\langle \tilde{\Theta} - \bar{\Theta}, \Theta - \bar{\Theta} \rangle. \end{aligned} \quad (\text{S31})$$

Taking conditional expectation given $\mathbf{y}$ yields

$$\mathbb{E}[\|\tilde{\Theta} - \Theta\|_2^2 \mid \mathbf{y}] = \|\tilde{\Theta} - \bar{\Theta}\|_2^2 + \mathbb{E}[\|\Theta - \bar{\Theta}\|_2^2 \mid \mathbf{y}], \quad (\text{S32})$$

because the cross term vanishes. The second term does not depend on $\tilde{\Theta}$, so the minimum is attained uniquely at $\tilde{\Theta} = \bar{\Theta}$. $\square$

Theorem 2. *The MAP estimator minimizes the negative log-posterior*

$$J(\Theta) := -\log \pi(\Theta \mid \mathbf{y}), \quad (\text{S33})$$

equivalently the penalized least-squares objective

$$J(\Theta) = \frac{1}{2}\|\mathbf{y} - K\Theta\|_{R^{-1}}^2 + \frac{1}{2}\|\Theta - \mathbf{m}_0\|_{\Sigma_0^{-1}}^2 + \text{const}. \quad (\text{S34})$$

Proof. This follows directly from (S24), since maximizing the posterior density is equivalent to minimizing its negative logarithm. $\square$

A.7 Induced posterior estimate for the state

The Stage 1 posterior state is

$$\mathbf{x}_t^{\text{prior}} = D(\mathbf{z}_t)\hat{\boldsymbol{\theta}}_t = \hat{\boldsymbol{\theta}}_t \odot \mathbf{z}_t. \quad (\text{S35})$$

Since $D(\mathbf{z}_t)$ is linear in $\boldsymbol{\theta}_t$ once $\mathbf{z}_t$ is fixed, if $\hat{\boldsymbol{\theta}}_t$ is the posterior mean then

$$\hat{\mathbf{x}}_t^{\text{prior}} = D(\mathbf{z}_t) \mathbb{E}[\boldsymbol{\theta}_t \mid \mathbf{y}] \quad (\text{S36})$$

is the corresponding posterior mean in the transformed parameterization. Therefore the Stage 1 state estimate is a proper posterior estimator induced by the Bayesian model.

A.8 What cannot be claimed: non-identifiability of the full five-layer state

Proposition 3. *The full five-layer hydrologic state is not uniquely identifiable from deformation alone without additional structure.*

Proof. The forward operator depends on the load-driven components only through the sum

$$S0_t + Ss_t + Sd_t + Sr_t. \quad (\text{S37})$$

Therefore any perturbation satisfying

$$\delta S0_t + \delta Ss_t + \delta Sd_t + \delta Sr_t = 0, \quad \delta Sg_t = 0, \quad (\text{S38})$$

lies in the null space of the load contribution and leaves the predicted deformation unchanged. Hence the inverse problem is non-injective on the full five-layer state space. $\square$

The mathematically correct statement is therefore not that the method recovers the unique physical truth, but that it yields a proper posterior and Bayes-optimal conditional estimates under the assumed model.

A.9 Constrained extension

Let $\mathbf{c}$ denote any additional observation or constraint data, for example external hydrologic information. Then the constrained posterior is

$$\pi(\Theta \mid \mathbf{y}, \mathbf{c}) \propto \pi(\mathbf{y} \mid \Theta) \pi(\mathbf{c} \mid \Theta) \pi(\Theta). \quad (\text{S39})$$

Equivalently, the negative log-posterior becomes

$$J_{\text{constr}}(\Theta) = J_{\text{pure}}(\Theta) + J_{\text{ext}}(\Theta). \quad (\text{S40})$$

Proposition 4. *If the additional likelihood $\pi(\mathbf{c} \mid \Theta)$ is proper Gaussian, or more generally if J_{ext} is proper, lower-semicontinuous, and coercive, then the constrained posterior remains proper and the same conclusions on posterior existence, MAP optimality, and Bayes-optimality under squared loss continue to hold.*

Proof. The additional term augments the posterior kernel by another proper penalty/likelihood factor. Under the stated regularity assumptions, the total negative log-posterior remains coercive, so the posterior remains proper. The MAP again minimizes the augmented negative log-posterior, and the posterior mean remains the Bayes estimator under squared loss. $\square$

A.10 Real-data conditioning and the grouped Bologna formulation

The previous results establish existence and Bayesian optimality under the assumed model, but they do not guarantee that every applied real-data discretization is numerically well-conditioned. In the Bologna case, some columns of the full layered real deformation design can become extremely small relative to the observed InSAR signal. The posterior may still exist, but the inferred coefficients can become numerically unstable or physically uninformative.

This motivates the grouped applied real-data state

$$\mathbf{x}_t^{\text{grp}} = \begin{bmatrix} \text{ShallowLoad}_t \\ \text{DeepLoad}_t \\ \text{Groundwater}_t \end{bmatrix} = \begin{bmatrix} S0_t + Ss_t \\ Sd_t + Sr_t \\ Sg_t \end{bmatrix}. \quad (\text{S41})$$

The rest of the Bayesian formulation then follows on a better-conditioned grouped state space. This does not change the mathematical structure of the inference problem; it changes only the applied parameterization. The grouped formulation is therefore interpreted as the practically supported real-data estimator for Bologna, while the unconstrained full layered inversion remains a theoretically meaningful but empirically unstable formulation in the current setting.

B Forward Hydro-Mechanical Formulation and Observation Operators

This appendix gives the forward hydro-mechanical formulation used to map hydrologic storage states to deformation-space observations and defines the observation operators used in the grouped multisensor Bologna inversion.

B.1 Full five-layer hydrologic state

At time t and spatial location $\mathbf{s} \in \Omega$, let the hydrologic state be

$$\mathbf{x}_t(\mathbf{s}) = \begin{bmatrix} S0_t(\mathbf{s}) \\ Ss_t(\mathbf{s}) \\ Sd_t(\mathbf{s}) \\ Sg_t(\mathbf{s}) \\ Sr_t(\mathbf{s}) \end{bmatrix}, \quad (\text{S42})$$

where $S0$, Ss , Sd , Sg , and Sr denote top-layer soil storage, shallow soil storage, deep soil storage, groundwater storage, and surface-water storage, respectively.

For the full deformation-space construction, the load-driven branch is formed from

$$\Delta\ell_t(\mathbf{s}) = \rho_w \left(S0_t(\mathbf{s}) + Ss_t(\mathbf{s}) + Sd_t(\mathbf{s}) + Sr_t(\mathbf{s}) \right), \quad (\text{S43})$$

while the groundwater-pressure branch is represented by

$$\Delta p_t(\mathbf{s}) = \rho_w g \frac{Sg_t(\mathbf{s})}{S_{\text{eff}}}, \quad (\text{S44})$$

where ρ_w is water density, g is gravitational acceleration, and S_{eff} is an effective storage parameter.

B.2 Continuous forward deformation model

Let G_{load} denote the elastic loading Green's operator and let G_{poro} denote the poroelastic groundwater-response operator. The load-driven and poroelastic displacements are

$$u_t^{\text{load}}(\mathbf{s}) = G_{\text{load}}[\Delta\ell_t](\mathbf{s}), \quad (\text{S45})$$

$$u_t^{\text{poro}}(\mathbf{s}) = G_{\text{poro}}[\Delta p_t](\mathbf{s}), \quad (\text{S46})$$

and the total displacement is

$$u_t^{\text{tot}}(\mathbf{s}) = u_t^{\text{load}}(\mathbf{s}) + u_t^{\text{poro}}(\mathbf{s}). \quad (\text{S47})$$

Equivalently, in kernel form,

$$u_t^{\text{load}}(\mathbf{s}) = \int_{\Omega} K_{\text{load}}(\mathbf{s}, \boldsymbol{\xi}) \Delta\ell_t(\boldsymbol{\xi}) d\boldsymbol{\xi}, \quad (\text{S48})$$

$$u_t^{\text{poro}}(\mathbf{s}) = \int_{\Omega} K_{\text{poro}}(\mathbf{s}, \boldsymbol{\xi}) \Delta p_t(\boldsymbol{\xi}) d\boldsymbol{\xi}, \quad (\text{S49})$$

where K_{load} and K_{poro} are the corresponding Green's kernels.

B.3 LOS projection and deformation observation

The observable InSAR signal is the line-of-sight (LOS) deformation

$$d_t(\mathbf{s}) = \mathcal{P}_{\text{LOS}}[u_t^{\text{tot}}](\mathbf{s}) + \varepsilon_t^{\text{I}}(\mathbf{s}), \quad (\text{S50})$$

where $\mathcal{P}_{\text{LOS}}(\cdot)$ denotes the radar projection operator and ε_t^{I} is the InSAR observation error. Under a dominant-vertical approximation,

$$d_t(\mathbf{s}) \approx \cos(\theta) u_{z,t}^{\text{tot}}(\mathbf{s}) + \varepsilon_t^{\text{I}}(\mathbf{s}), \quad (\text{S51})$$

with incidence angle θ and vertical displacement component $u_{z,t}^{\text{tot}}$.

B.4 Discrete operator form

After discretization on a grid with P cells, all fields become vectors in $\mathbb{R}^P$. Let

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{x}_t^{(S0)} \\ \mathbf{x}_t^{(Ss)} \\ \mathbf{x}_t^{(Sd)} \\ \mathbf{x}_t^{(Sg)} \\ \mathbf{x}_t^{(Sr)} \end{bmatrix} \in \mathbb{R}^{5P}, \quad (\text{S52})$$

where each block belongs to $\mathbb{R}^P$.

Define the discrete loading and poroelastic operators

$$A_{\text{load}} \in \mathbb{R}^{P \times P}, \quad A_{\text{poro}} \in \mathbb{R}^{P \times P}, \quad (\text{S53})$$

and the LOS projection matrix

$$P_{\text{LOS}} \in \mathbb{R}^{P \times P}. \quad (\text{S54})$$

Then the full discrete deformation model is

$$\mathbf{d}_t = P_{\text{LOS}} \left[A_{\text{load}} \left(\mathbf{x}_t^{(S0)} + \mathbf{x}_t^{(Ss)} + \mathbf{x}_t^{(Sd)} + \mathbf{x}_t^{(Sr)} \right) + A_{\text{poro}} \mathbf{x}_t^{(Sg)} \right] + \varepsilon_t^{\text{I}}. \quad (\text{S55})$$

For the deformation-space Bayesian construction, one may also define component-wise deformation predictors,

$$\mathbf{u}_t^{(S0)} = P_{\text{LOS}} A_{\text{load}} \mathbf{x}_t^{(S0)}, \quad \mathbf{u}_t^{(Ss)} = P_{\text{LOS}} A_{\text{load}} \mathbf{x}_t^{(Ss)}, \quad (\text{S56})$$

$$\mathbf{u}_t^{(Sd)} = P_{\text{LOS}} A_{\text{load}} \mathbf{x}_t^{(Sd)}, \quad \mathbf{u}_t^{(Sr)} = P_{\text{LOS}} A_{\text{load}} \mathbf{x}_t^{(Sr)}, \quad (\text{S57})$$

$$\mathbf{u}_t^{(Sg)} = P_{\text{LOS}} A_{\text{poro}} \mathbf{x}_t^{(Sg)}. \quad (\text{S58})$$

At each grid cell p , the corresponding deformation-side predictor vector is

$$\mathbf{Z}_{t,p} = \begin{bmatrix} u_{t,p}^{(S0)} & u_{t,p}^{(Ss)} & u_{t,p}^{(Sd)} & u_{t,p}^{(Sg)} & u_{t,p}^{(Sr)} \end{bmatrix}. \quad (\text{S59})$$

This is the design used by the full deformation-space Stage 1 Bayesian model.

B.5 Grouped state used in the applied Bologna inversion

Because the unconstrained full five-layer real-data inversion was found to be poorly conditioned in the Bologna case, the applied real-data estimator uses the grouped state

$$\mathbf{x}_t^{\text{grp}} = \begin{bmatrix} \text{ShallowLoad}_t \\ \text{DeepLoad}_t \\ \text{Groundwater}_t \end{bmatrix} = \begin{bmatrix} S0_t + Ss_t \\ Sd_t + Sr_t \\ Sg_t \end{bmatrix}, \quad (\text{S60})$$

where

$$\text{ShallowLoad}_t \in \mathbb{R}^P, \quad \text{DeepLoad}_t \in \mathbb{R}^P, \quad \text{Groundwater}_t \in \mathbb{R}^P. \quad (\text{S61})$$

The grouped forward deformation model is therefore

$$\mathbf{d}_t = P_{\text{LOS}} [A_{\text{load}}(\text{ShallowLoad}_t + \text{DeepLoad}_t) + A_{\text{poro}}\text{Groundwater}_t] + \boldsymbol{\varepsilon}_t^{\text{I}}. \quad (\text{S62})$$

In compact operator form,

$$\mathbf{d}_t = H_t^{\text{I}} \mathbf{x}_t^{\text{grp}} + \boldsymbol{\varepsilon}_t^{\text{I}}, \quad (\text{S63})$$

with

$$H_t^{\text{I}} = P_{\text{LOS}} \begin{bmatrix} A_{\text{load}} & A_{\text{load}} & A_{\text{poro}} \end{bmatrix} \in \mathbb{R}^{P \times 3P}. \quad (\text{S64})$$

B.6 Regional aggregation operators

Several external observations enter the grouped multisensor inversion only after spatial aggregation over the Bologna overlap domain. Let

$$M \in \mathbb{R}^{1 \times P} \quad (\text{S65})$$

denote a regional averaging operator over the shared Bologna domain. Then for any gridded state variable $\mathbf{v}_t \in \mathbb{R}^P$, the corresponding regional mean is

$$\bar{v}_t = M \mathbf{v}_t. \quad (\text{S66})$$

B.7 GRACE observation operator

GRACE provides a coarse regional total-storage-type anomaly. In the grouped formulation, the corresponding observation equation is

$$y_t^{\text{G}} = H_t^{\text{G}} \mathbf{x}_t^{\text{grp}} + \varepsilon_t^{\text{G}}, \quad (\text{S67})$$

where

$$H_t^{\text{G}} = \begin{bmatrix} M & M & M \end{bmatrix} \in \mathbb{R}^{1 \times 3P}, \quad (\text{S68})$$

and

$$\varepsilon_t^{\text{G}} \sim \mathcal{N}(0, R_t^{\text{G}}). \quad (\text{S69})$$

Thus the grouped state is linked to a regional total-storage anomaly through the sum of the grouped storage components.

B.8 SMAP observation operator

SMAP is used as an external shallow hydrologic constraint. In the grouped formulation, the simplest observation equation is

$$y_t^S = H_t^S \mathbf{x}_t^{\text{grp}} + \varepsilon_t^S, \quad (\text{S70})$$

with

$$H_t^S = \begin{bmatrix} c_S M & 0 & 0 \end{bmatrix} \in \mathbb{R}^{1 \times 3P}, \quad (\text{S71})$$

where c_S is a scaling factor converting the grouped shallow-load state into the SMAP-compatible anomaly space, and

$$\varepsilon_t^S \sim \mathcal{N}(0, R_t^S). \quad (\text{S72})$$

More elaborate observation operators can be used if a different mapping between grouped storage and retrieved surface soil moisture is desired.

B.9 SWOT observation operator

SWOT is treated as an exploratory external surface-water constraint. Because the grouped state does not isolate Sr separately, SWOT enters through a grouped approximation,

$$y_t^W = H_t^W \mathbf{x}_t^{\text{grp}} + \varepsilon_t^W, \quad (\text{S73})$$

with

$$H_t^W = \begin{bmatrix} 0 & c_W M & 0 \end{bmatrix} \in \mathbb{R}^{1 \times 3P}, \quad (\text{S74})$$

where c_W is an effective scaling coefficient reflecting the fact that the grouped deep-load state contains the surface-water contribution through the $Sd + Sr$ block. The SWOT error term is

$$\varepsilon_t^W \sim \mathcal{N}(0, R_t^W). \quad (\text{S75})$$

In the current Bologna experiments, SWOT is technically integrated but remains sparse in time and therefore acts only as an exploratory added constraint rather than a major driver of the final result.

B.10 Grouped multisensor observation model

The grouped multisensor Bologna inversion can therefore be written in stacked form as

$$\mathbf{y}_t = \begin{bmatrix} \mathbf{d}_t \\ y_t^G \\ y_t^S \\ y_t^W \end{bmatrix} = \begin{bmatrix} H_t^I \\ H_t^G \\ H_t^S \\ H_t^W \end{bmatrix} \mathbf{x}_t^{\text{grp}} + \begin{bmatrix} \varepsilon_t^I \\ \varepsilon_t^G \\ \varepsilon_t^S \\ \varepsilon_t^W \end{bmatrix}. \quad (\text{S76})$$

Equivalently,

$$\mathbf{y}_t = H_t \mathbf{x}_t^{\text{grp}} + \boldsymbol{\varepsilon}_t, \quad (\text{S77})$$

where H_t is the full grouped multisensor observation operator and $\boldsymbol{\varepsilon}_t$ is the corresponding stacked error vector.

B.11 State evolution in the grouped Bologna model

The applied grouped state evolves according to a linear stochastic dynamics model,

$$\mathbf{x}_t^{\text{grp}} = A \mathbf{x}_{t-1}^{\text{grp}} + \boldsymbol{\eta}_t, \quad (\text{S78})$$

where $A \in \mathbb{R}^{3P \times 3P}$ is the grouped transition operator and

$$\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, Q_t). \quad (\text{S79})$$

In practice, the grouped balanced multisensor Stage 1 model corrects the grouped W3RA prior state under this multisensor observation system and provides the current best supported Bologna estimator.

B.12 Interpretation

The equations above define two related operator families:

1. the full five-layer deformation-space operator used to formulate the general inversion framework and the synthetic validation problem;
2. the grouped three-state multisensor operator used in the practically supported Bologna inversion.

The first is the theoretically richer model. The second is the currently validated real-data formulation because it yields physically plausible grouped posteriors, stable internal multisensor fit, and meaningful independent support from groundwater wells.

C Conditioning Diagnostics of the Bologna Inversion

This appendix formalizes the conditioning diagnostics used to explain why the original real-data layered inversion was ill-conditioned and how the grouped balanced multisensor Stage 1 improved the inverse problem. The diagnostics were computed from the saved runs using the conditioning workflow and summarized in the corresponding JSON outputs.

C.1 Purpose

The role of this appendix is to quantify, in operator form, the difference between:

1. the earlier unconstrained layered real-data deformation-space inversion,
2. the earlier grouped patch formulation, and
3. the current grouped balanced multisensor Stage 1 formulation.

The key question is whether the practical shift to the grouped Bologna model is justified only heuristically, or whether it is also supported by explicit conditioning diagnostics. The results below show that the grouped balanced multisensor formulation improves the inverse problem by several orders of magnitude in conditioning and restores practical rank.

C.2 Inverse-problem setting

C.2.1 Old deformation-space Stage 1 formulation

The original real-data deformation-space formulation may be written schematically as

$$Y_{t,p} = Z_{t,p} \theta_{t,p} + \varepsilon_{t,p}, \quad (\text{S80})$$

where $Y_{t,p}$ is observed deformation, $Z_{t,p}$ is the forward-converted W3RA predictor in deformation space, and $\theta_{t,p}$ is a latent correction coefficient.

In the full layered case,

$$Z_{t,p} = \begin{bmatrix} u_{t,p}^{(S0)} & u_{t,p}^{(Ss)} & u_{t,p}^{(Sd)} & u_{t,p}^{(Sg)} & u_{t,p}^{(Sr)} \end{bmatrix}, \quad (\text{S81})$$

while in the earlier grouped patch case,

$$Z_{t,p}^{\text{grp}} = \begin{bmatrix} u_{t,p}^{(\text{LoadTotal})} & u_{t,p}^{(Sg)} \end{bmatrix}. \quad (\text{S82})$$

To analyze conditioning, all valid space-time rows were stacked into a single design matrix,

$$J_{\text{stack}} = \begin{bmatrix} Z_{1,1} \\ Z_{1,2} \\ \vdots \\ Z_{T,P} \end{bmatrix}. \quad (\text{S83})$$

C.2.2 Current grouped balanced multisensor Stage 1

The current real-data grouped Stage 1 uses the grouped state

$$x_t = \begin{bmatrix} \text{ShallowLoad}_t \\ \text{DeepLoad}_t \\ \text{Groundwater}_t \end{bmatrix} = \begin{bmatrix} S0_t + Ss_t \\ Sd_t + Sr_t \\ Sg_t \end{bmatrix}, \quad (\text{S84})$$

with dynamic correction factors

$$\theta_t = \phi\theta_{t-1} + (1 - \phi)\theta_{\text{ref}} + \eta_t, \quad x_t = \theta_t \odot z_t. \quad (\text{S85})$$

At each time t , the available observations are represented by one or more rows of a balanced observation operator H_t .

InSAR row

$$h_t^{(\text{InSAR})} = \frac{z_t^{(\text{def})}}{s_{\text{insar}}}, \quad (\text{S86})$$

where $z_t^{(\text{def})}$ is the grouped deformation-side predictor and s_{insar} is the balancing scale.

GRACE row

$$h_t^{(\text{GRACE})} = \frac{z_t^{(\text{state})}}{s_{\text{grace}}}, \quad (\text{S87})$$

where $z_t^{(\text{state})}$ is the grouped state prior.

SMAP row

$$h_t^{(\text{SMAP})} = \frac{1}{s_{\text{smap}}} \begin{bmatrix} \beta_{\text{smap}} z_{t,\text{shallow}} & 0 & 0 \end{bmatrix}. \quad (\text{S88})$$

SWOT row

$$h_t^{(\text{SWOT})} = \frac{1}{s_{\text{swot}}} \begin{bmatrix} \beta_{\text{swot}} z_{t,\text{shallow}} & 0 & 0 \end{bmatrix}. \quad (\text{S89})$$

All valid rows across all times were stacked as

$$J_{\text{stack}} = \begin{bmatrix} H_1 \\ H_2 \\ \vdots \\ H_T \end{bmatrix}. \quad (\text{S90})$$

C.3 Conditioning metrics

C.3.1 Column-scale spread

For column k of the stacked operator J , define

$$c_k = \|J_{:,k}\|_2. \quad (\text{S91})$$

The column-scale spread is

$$\text{spread}(J) = \frac{\max_k c_k}{\min_k c_k}. \quad (\text{S92})$$

This directly measures scale imbalance. Large spread indicates that some state directions are much more visible to the observations than others.

The use of matrix norms, singular values, condition numbers, and numerical rank follows standard numerical linear algebra treatments such as [1, 2].

C.3.2 Singular values and operator condition number

Let the singular values of J be

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0. \quad (\text{S93})$$

The operator condition number is

$$\kappa(J) = \frac{\sigma_{\max}(J)}{\sigma_{\min}(J)} = \frac{\sigma_1}{\sigma_r}. \quad (\text{S94})$$

Large $\kappa(J)$ indicates high sensitivity of the inverse problem to perturbations in the data or model [1, 2].

C.3.3 Thresholded numerical rank

We used a thresholded singular-value rank,

$$r_{\text{num}}(\tau) = \# \{i : \sigma_i > \tau \sigma_1\}, \quad (\text{S95})$$

with $\tau = 10^{-6}$ and $\tau = 10^{-4}$. If r_{num} is smaller than the nominal state dimension, then one or more state directions are practically lost.

C.3.4 Information matrix / Gauss–Newton Hessian

For a least-squares or Gaussian likelihood,

$$\mathcal{L}(x) = \frac{1}{2}(y - h(x))^{\top} R^{-1}(y - h(x)). \quad (\text{S96})$$

Linearizing around a reference state gives the Gauss–Newton approximation

$$H_{\text{GN}} \approx J^{\top} R^{-1} J. \quad (\text{S97})$$

This same matrix is the data-information matrix

$$\mathcal{I} = J^{\top} R^{-1} J. \quad (\text{S98})$$

In the old static diagnostics we used the unweighted version $J^{\top} J$ to diagnose raw operator conditioning. In the grouped state-space model we used the sensor-weighted sum

$$\mathcal{I} = \sum_{t,m} \frac{1}{r_m^2} h_t^{(m)\top} h_t^{(m)}, \quad (\text{S99})$$

where m indexes the available sensors at time t . This is the natural Hessian-like information matrix for the balanced grouped problem [4].

Let the eigenvalues of $\mathcal{I}$ be

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n. \quad (\text{S100})$$

Then the information-matrix condition number is

$$\kappa(\mathcal{I}) = \frac{\lambda_{\max}}{\lambda_{\min}}. \quad (\text{S101})$$

Since $\mathcal{I} = J^{\top} J$ in the simplest case,

$$\kappa(\mathcal{I}) \approx \kappa(J)^2. \quad (\text{S102})$$

C.3.5 Column correlation / separability

For columns i and j of J , we computed

$$\rho_{ij} = \text{corr}(J_{:,i}, J_{:,j}). \quad (\text{S103})$$

If $|\rho_{ij}|$ is close to 1, then the corresponding state directions generate very similar observation responses and are difficult to separate. This is not itself a condition number, but it is a direct practical separability diagnostic for layered or grouped states.

C.3.6 Observability Gramian

For the dynamic state-space model

$$x_t = Ax_{t-1} + \eta_t, \quad y_t = H_t x_t + \epsilon_t, \quad (\text{S104})$$

the observability Gramian over the full time window is

$$W_o = \sum_{t=1}^T \Phi_{t,0}^\top H_t^\top R_t^{-1} H_t \Phi_{t,0}, \quad (\text{S105})$$

where $\Phi_{t,0}$ is the state transition from the initial time to time t . This is the standard observability construction in linear systems theory [3].

In the grouped Kalman formulation,

$$A = \phi I, \quad (\text{S106})$$

so

$$\Phi_{t,0} = \phi^t I. \quad (\text{S107})$$

Therefore the Gramian reduces to

$$W_o = \sum_{t=1}^T \phi^{2(T-t)} H_t^\top R_t^{-1} H_t. \quad (\text{S108})$$

Its condition number is

$$\kappa(W_o) = \frac{\lambda_{\max}(W_o)}{\lambda_{\min}(W_o)}. \quad (\text{S109})$$

This metric quantifies whether the grouped state directions remain dynamically observable over the full assimilation window.

C.4 How the diagnostics were computed from the saved runs

C.4.1 Old layered and grouped patch runs

For the old Stage 1 patch runs:

1. the saved W3RA patch state cube $z_{\text{layers}}(t, k, y, x)$ was loaded;
2. each layer was forward-converted to deformation-space components using the same elastic and poroelastic physics used in the inversion;
3. the resulting component responses were stacked over all valid (t, y, x) rows;
4. the diagnostics were computed twice:
 - (a) on the raw operator, to quantify the actual physical scaling problem,
 - (b) on the column-standardized operator, to separate scale imbalance from pure collinearity.

C.4.2 Current grouped multisensor Stage 1

For the current grouped Kalman model:

1. the balanced observation rows used in the regional grouped inversion were rebuilt:

$$h_t^{(\text{InSAR})}, h_t^{(\text{GRACE})}, h_t^{(\text{SMAP})}, h_t^{(\text{SWOT})}; \quad (\text{S110})$$

2. all valid rows across time were stacked to form J_{stack} ;

3. the following were then computed:

- (a) singular values and $\kappa(J_{\text{stack}})$,
- (b) the weighted information matrix $\mathcal{I}$,
- (c) the observability Gramian W_o .

C.5 Result tables

Table S1: Raw-operator conditioning before and after grouping and multisensor balancing.

Formulation	$\kappa(J)$	$\kappa(\mathcal{I})$	numerical rank	scale spread
Old 5-layer patch	2.16×10^6	8.81×10^{10}	4/5	1.26×10^6
Old grouped patch	4.54×10^4	2.06×10^9	2/2	3.01×10^4
Current grouped multisensor Kalman	6.99×10^1	4.88×10^3	3/3	6.70×10^1

Table S2: Column-standardized static operators. This shows that the old five-layer failure was driven primarily by extreme scale imbalance, with additional correlation structure on top.

Formulation	$\kappa(J_{\text{std}})$	$\kappa(\mathcal{I}_{\text{std}})$	numerical rank
Old 5-layer patch	5.09	25.88	5/5
Old grouped patch	2.63	6.93	2/2

Table S3: Separability and dynamic observability diagnostics.

Formulation	$\max \rho_{ij} $ off diagonal	$\kappa(W_o)$
Old 5-layer patch	0.808	—
Old grouped patch	0.749	—
Current grouped multisensor Kalman	0.259	2.12×10^4

C.6 Interpretation

These diagnostics support the following conclusions.

1. The original real-data layered deformation-space inversion was genuinely ill-conditioned, not merely inconvenient. Its raw stacked sensitivity operator had extreme column-scale disparity, a very large condition number, and a loss of practical rank.
2. Grouping the state reduced the problem substantially, but the old grouped patch formulation was still poorly scaled because the load-related response remained much smaller than the groundwater-related response.

3. The current grouped balanced multisensor Stage 1 improved the problem in three main ways:
 - (a) the operator condition number dropped by several orders of magnitude,
 - (b) the grouped state directions became fully ranked,
 - (c) the inter-column correlations became much smaller, so the grouped state directions are materially more separable.
4. The observability Gramian of the current grouped model is not perfectly isotropic, but it is positive in all three grouped directions, showing that the balanced multisensor state is dynamically observable over the assimilation window.

In summary, the conditioning diagnostics quantitatively support the practical shift from an unconstrained real layered inversion to the current grouped balanced multisensor Stage 1 Bologna formulation.

D Synthetic Validation and External Well-Validation Details

This appendix documents the synthetic validation protocol used to test the inversion machinery under controlled conditions and the external groundwater-well validation used to assess the grouped Bologna posterior against independent observations. The appendix complements the main paper by providing the detailed setup, quantitative summaries, and placeholder locations for the extended figures that support the interpretation of the real-data results.

D.1 Synthetic validation protocol

Before applying the inversion to the Bologna overlap, the method was tested in a controlled synthetic setting in which the latent hydrologic state is known. The purpose of this pre-stage was not to claim that synthetic success guarantees real-data identifiability, but to verify that the deformation-space Bayesian machinery can recover known signals under model-consistent conditions.

The synthetic workflow consisted of:

1. generating smooth W3RA-like storage fields for the five components

$$\{S0, Ss, Sd, Sg, Sr\},$$

2. converting these storage fields into deformation-space predictors using the same forward hydro-mechanical operators used in the real formulation,
3. constructing synthetic observations of the form

$$Y_{t,p} = Z_{t,p}\theta_{t,p} + \varepsilon_{t,p}, \tag{S111}$$

4. running Stage 1 Bayesian recovery to obtain posterior states, and
5. testing whether Stage 2 residual learning can improve the Stage 1 posterior when the true residual is known.

The principal clean validation setup used:

- 24 time steps,
- a 12×12 spatial grid,

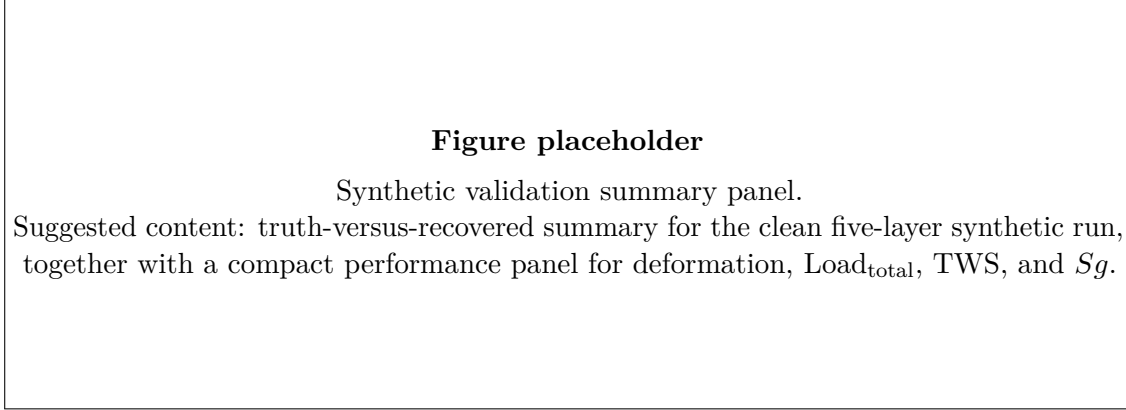


Figure S1: Synthetic validation summary. This figure should show the main clean-case synthetic feasibility result that justifies the deformation-space inversion machinery before real-data application.

- the full five-layer synthetic state,
- and `noise_scale = 0.0`.

As summarized in Table S4, the clean synthetic run showed strong state recovery across the key aggregated quantities and for groundwater. In particular, the clean Stage 1 synthetic run reached approximately $R^2 \approx 0.976$ for Load_{total}, $R^2 \approx 0.980$ for total water storage, $R^2 \approx 0.980$ for deformation reconstruction, and $R^2 \approx 0.981$ for *Sg*. These results justify the use of the synthetic block as a pre-stage feasibility test for the inversion machinery.

Table S4: Clean synthetic Stage 1 validation summary used as the main synthetic feasibility anchor.

Quantity	Metric	Value
Load _{total} state	R^2	0.976
TWS state	R^2	0.980
Deformation reconstruction	R^2	0.980
<i>Sg</i> state	R^2	0.981

As shown conceptually in Figure S1, the clean synthetic case is the correct anchor for evaluating whether the inverse machinery works under model-consistent conditions. The noisy synthetic variants remain useful as stress tests, but the current paper uses the clean benchmark as the main synthetic evidence because the real applied contribution of the study lies in the grouped Bologna result rather than in a noise-robust learned benchmark.

D.2 External well-validation methodology

The strongest external validation available for the grouped Bologna posterior is the independent groundwater-well network. Since the grouped posterior is a storage-like latent estimate while the wells measure hydraulic head, validation was carried out in anomaly space rather than by direct equality of units.

The well-validation workflow was:

1. interpolate the grouped posterior fields from the Kalman tile/grid support to each well location,
2. align the posterior and well series by nearest date,

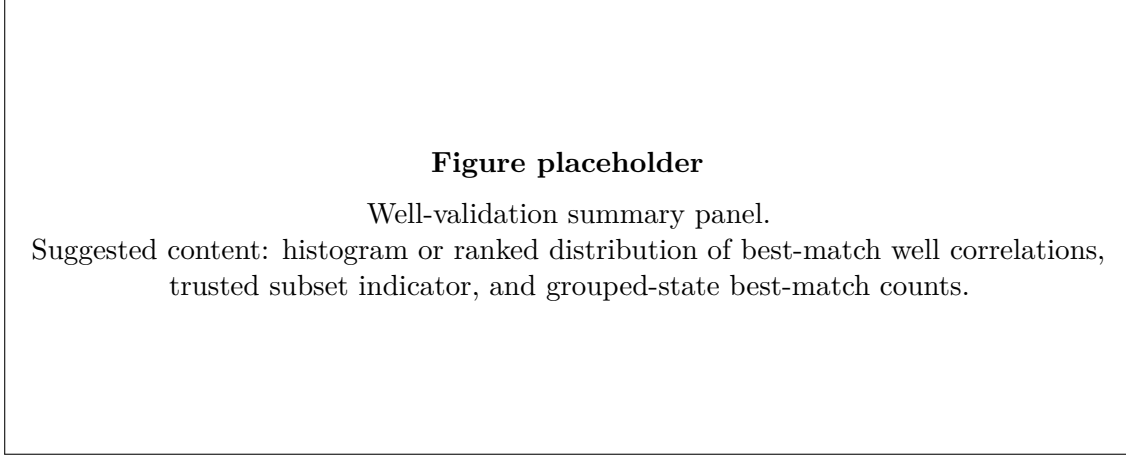


Figure S2: Extended well-validation summary panel for the grouped Bologna posterior.

3. compute anomaly time series,
4. search over a discrete lag set

$$\mathcal{L}_{\text{lag}} = \{-90, -60, -30, -14, 0, 14, 30, 60, 90\} \text{ days}, \quad (\text{S112})$$

5. and record the best state / best lag match for each station.

This lag-aware anomaly validation is important because the grouped posterior and the well-head response need not be synchronous at all locations. It also allows the grouped state interpretation to remain physically reasonable in a coupled deformation system in which different grouped components may dominate at different stations.

D.3 Overall well-validation results

The overall validation summary is given in [Table S5](#). A total of 83 station series were evaluated. After best-state and best-lag matching, all 83 stations showed positive correlation, 74 stations had correlation at least 0.3, and 62 stations had correlation at least 0.5. The median anomaly correlation was approximately 0.704, while the top-10 mean correlation was approximately 0.947.

Table S5: Overall grouped-posterior well-validation summary.

Validation statistic	Value
Evaluated station series	83
Positive best-match correlations	83
Stations with correlation ≥ 0.3	74
Stations with correlation ≥ 0.5	62
Median best-match correlation	0.704
Top-10 mean best-match correlation	0.947
Trusted stations	21
Trusted hydrogeologic groups	6

The overall distribution of well-validation performance should be visualized in [Figure S2](#). That panel is intended to serve as the supplementary counterpart of the main paper's shorter well-validation figure and should summarize the correlation distribution, the trusted subset, and the best-state counts.

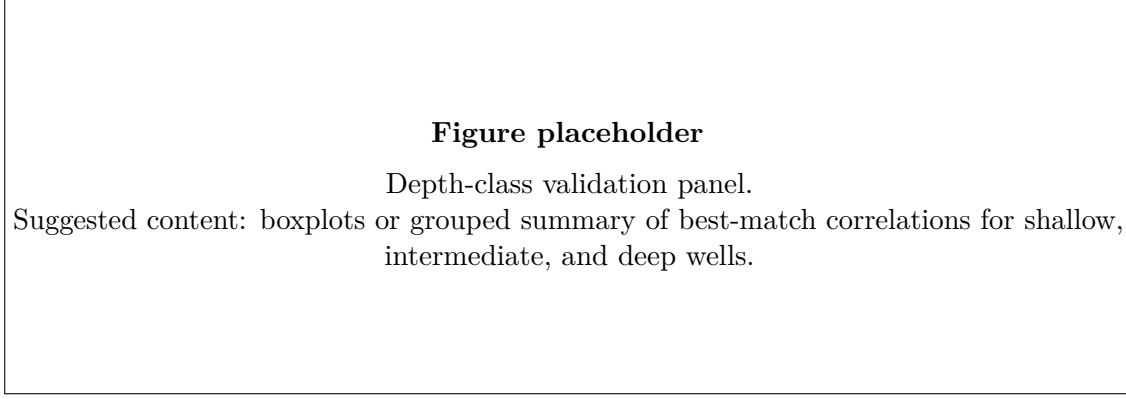


Figure S3: Depth-stratified well-validation performance for the grouped Bologna posterior.

D.4 Depth dependence and best-state interpretation

The depth-stratified summary is reported in [Table S6](#). The grouped posterior is supported across depth classes, with the strongest median anomaly correlations in the shallow and intermediate ranges. This pattern is physically plausible because shallower and intermediate wells are more likely to respond directly to storage changes reflected in the grouped posterior at the spatial and temporal scales considered here.

In addition, the best-state counts show that the grouped groundwater component is the best match for the largest number of stations, but not for all stations. This is also physically reasonable: in a coupled hydro-mechanical system, the observed local behavior can reflect not only grouped groundwater but also loading-related grouped states.

Table S6: Depth-stratified validation and best-state interpretation for the grouped Bologna posterior.

Category	Statistic	Value
Shallow wells	median correlation	0.767
Intermediate wells	median correlation	0.731
Deep wells	median correlation	0.617
Best state = Groundwater	station count	42
Best state = ShallowLoad	station count	25
Best state = DeepLoad	station count	16

The depth-class breakdown can be illustrated in [Figure S3](#), while representative station-level comparisons can be shown in [Figure S4](#). These figures are especially useful for supplementary interpretation because they provide local examples without overloading the main paper.

D.5 Trusted validation subset

To avoid over-interpreting hydrogeologic settings with weaker support, a conservative trusted subset was defined. The trusted subset contains 21 stations across 6 hydrogeologic groups, listed in [Table S7](#). This subset is the strongest external-validation panel and should be used for the most conservative interpretation of the grouped groundwater signal.

The spatial distribution of the trusted subset should be shown in [Figure S5](#). This figure should overlay the trusted stations on the Bologna inversion domain and, if useful, distinguish the trusted hydrogeologic groups by symbol or color.

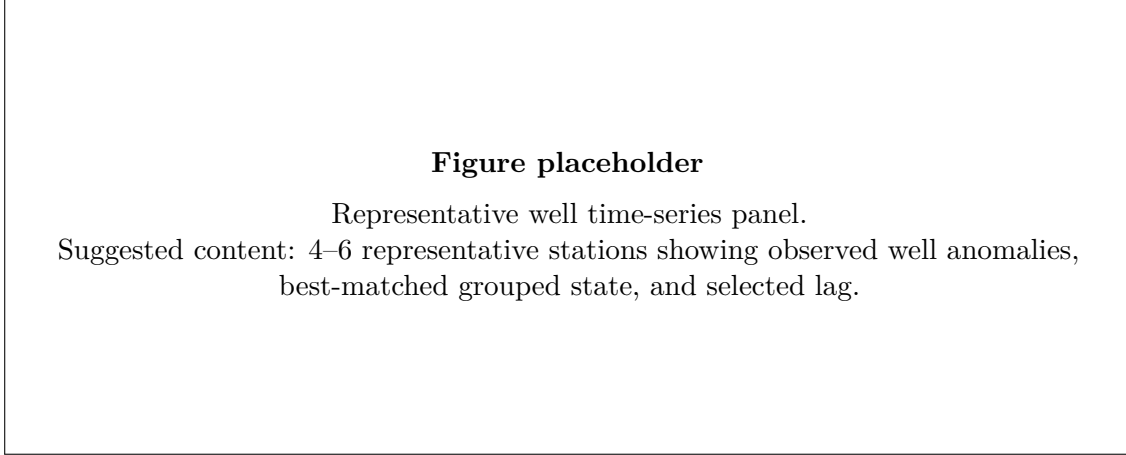


Figure S4: Representative station-level anomaly comparisons between the grouped posterior and independent wells.

Table S7: Trusted hydrogeologic groups used for the conservative external-validation subset.

Index	Trusted hydrogeologic group
1	Pianura Alluvionale Appenninica - confinato superiore
2	Freatico di pianura fluviale
3	Conoide Zena-Idice - confinato superiore
4	Conoide Reno-Lavino - libero
5	Conoide Reno-Lavino - confinato inferiore
6	Conoidi montane e Sabbie gialle orientali

D.6 Interpretation

The synthetic and well-validation results in this appendix support the following interpretation.

First, the synthetic validation block demonstrates that the inversion machinery is internally coherent and able to recover known signals in a model-consistent setting, as summarized in [Table S4](#) and [figure S1](#). Second, the external well-validation results in [Tables S5](#) to [S7](#) show that the grouped Bologna posterior is not merely reproducing InSAR internally, but is also consistent with independent groundwater behavior over a substantial fraction of the well network. Third, the trusted subset provides a conservative basis for interpretation and should therefore be preferred whenever the paper needs to make the strongest external-validation statement.

In summary, the current external evidence supports the grouped balanced multisensor Stage 1 model as the strongest real-data estimator in the Bologna application, while the earlier unconstrained full layered inversion remains unsupported and the current Stage 2 Swin refinement remains exploratory.

References

- [1] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4 edition, 2013.
- [2] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, Philadelphia, 2 edition, 2002. doi: 10.1137/1.9780898718027.
- [3] Rudolf E. Kalman. On the general theory of control systems. In *Proceedings of the First International Congress on Automatic Control*, pages 481–492, London, 1960. Butterworths.

Figure placeholder

Trusted wells map.

Suggested content: Bologna overlap domain, all wells, trusted wells highlighted, and optional hydrogeologic-group symbols.

Figure S5: Trusted external-validation subset for the grouped Bologna posterior.

- [4] Andrew M. Stuart. Inverse problems: A Bayesian perspective. *Acta Numerica*, 19:451–559, 2010. doi: 10.1017/S0962492910000061.